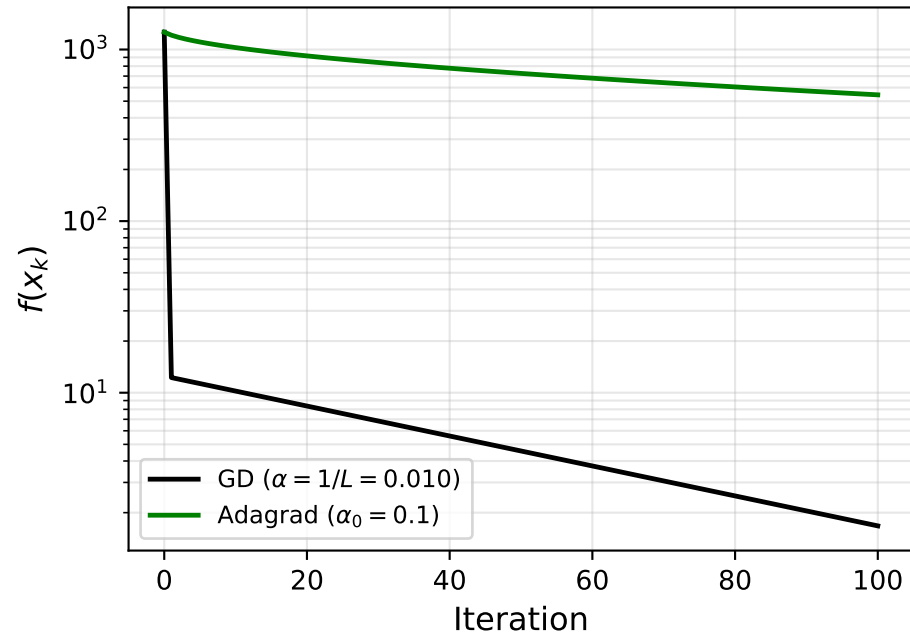
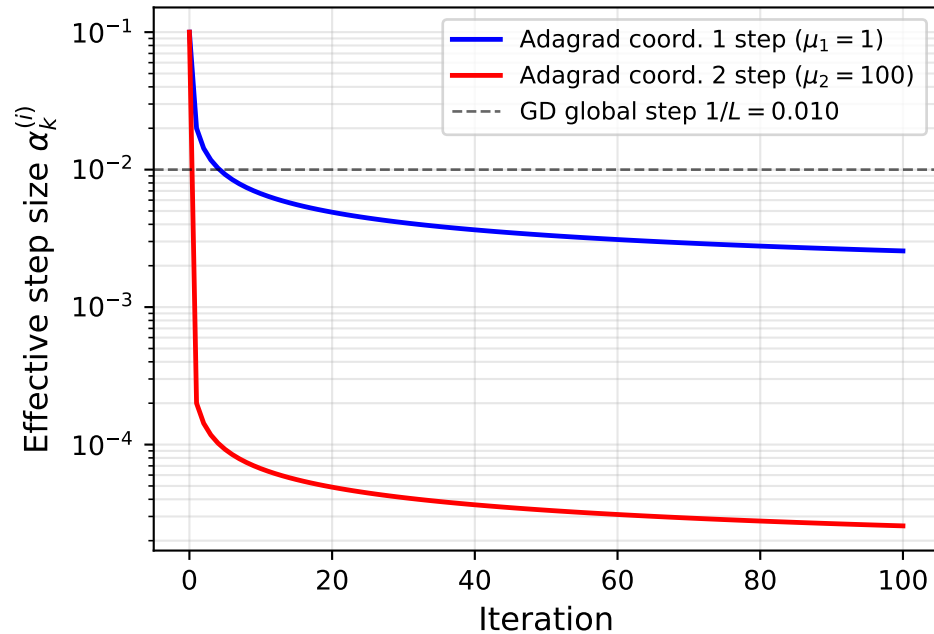


Convergence: Adagrad vs GD
(anisotropic $\mu_1 = 1, \mu_2 = 100$)



Per-coordinate Adagrad step sizes



Theoretical $O(1/\sqrt{k})$ vs empirical decay

